

Extra Paper

Clearly label which question you are answering.

Question 1 (21 Marks):

Question	1	2	3	4	5	6	7
Answer	b	a	b	a	b	a	b

Question 2 (15 Marks):

Q2.(a) Solution:

Part 1: Summation Notation

(5 Marks)

$$\text{Total ancestors from level 1 to } n = \sum_{k=1}^n 2^k$$

Part 2: Proof using Geometric Series Formula

(10 Marks)

We wish to prove that $\sum_{k=1}^n 2^k = 2^{n+1} - 2$.

Step 1 — Expand the sum:

$$\sum_{k=1}^n 2^k = 2^1 + 2^2 + 2^3 + \cdots + 2^n$$

Step 2 — Identify the geometric series parameters:

This is a geometric series with:

- First term: $a = 2$
- Common ratio: $r = 2$
- Number of terms: n

Step 3 — Apply the geometric series formula:

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{2(2^n - 1)}{2 - 1} = \frac{2 \cdot (2^n - 1)}{1} = 2 \cdot 2^n - 2$$

Step 4 — Simplify:

$$S_n = 2^{n+1} - 2 \quad \blacksquare$$

Question 3 (35 Marks):

Q3.(a) Prove that $\frac{(2n)!}{(2(n-1))!} = 4n^2 - 2n$ for any integer $n \geq 1$.

(10 Marks)

Step 1 — Rewrite the denominator:

$$2(n-1) = 2n - 2$$

So the expression becomes:

$$\frac{(2n)!}{(2n-2)!}$$

Step 2 — Expand $(2n)!$ recursively down to $(2n-2)!$:

By the recursive definition of factorial: $m! = m \cdot (m - 1)!$, applied twice:

$$\begin{aligned}(2n)! &= (2n) \cdot (2n - 1)! \\ &= (2n) \cdot (2n - 1) \cdot (2n - 2)!\end{aligned}$$

Step 3 — Cancel $(2n - 2)!$:

$$\frac{(2n)!}{(2n - 2)!} = \frac{(2n)(2n - 1)(2n - 2)!}{(2n - 2)!} = (2n)(2n - 1)$$

Step 4 — Expand the product:

$$(2n)(2n - 1) = 4n^2 - 2n$$

Conclusion:

$$\boxed{\frac{(2n)!}{(2(n - 1))!} = 4n^2 - 2n} \quad \blacksquare$$

Q3.(b) Set Operations on A, B, C .

(25 Marks)

Given:

$$A = \{x \in \mathbb{Z} \mid 1 \leq x \leq 6\} = \{1, 2, 3, 4, 5, 6\}$$

$$B = \{x \in A \mid x \text{ is even}\}$$

$$C = \{x \in \mathbb{Z} \mid x = 2k + 1 \text{ for some integer } k\} \quad (\text{odd integers})$$

Part 1: Roster form of B and $A \cap C$

(5 Marks)

$$B = \{2, 4, 6\}$$

$A \cap C$ = elements of A that are odd integers:

$$A \cap C = \{1, 3, 5\}$$

Part 2: Set Difference $A - B$ and its relationship to B^c

(10 Marks)

$$A - B = \{x \in A \mid x \notin B\} = \{1, 3, 5\}$$

The complement of B relative to the universal set $\mathbb{Z}$ is:

$$B^c = \{x \in \mathbb{Z} \mid x \notin B\} = \{\dots, -3, -2, -1, 0, 1, 3, 5, 7, \dots\} \quad (\text{all non-even integers})$$

Relationship:

$$A - B = A \cap B^c$$

That is, the set difference equals the intersection of A with the complement of B . In roster form: $A - B = \{1, 3, 5\} = A \cap B^c \cap A = A \cap B^c$.

Part 3: Does $\mathcal{H} = \{B, (A - B)\}$ form a valid partition of A ?

(10 Marks)

We verify the three conditions of a partition of $A = \{1, 2, 3, 4, 5, 6\}$:

1) **Non-emptiness:**

- $B = \{2, 4, 6\} \neq \emptyset \quad \checkmark$
- $A - B = \{1, 3, 5\} \neq \emptyset \quad \checkmark$

2) Pairwise Disjointness:

$$B \cap (A - B) = \emptyset$$

By definition, $A - B$ consists of elements in A but *not* in B . So no element can belong to both B and $A - B$. ✓

3) Exhaustiveness (Union equals A):

$$B \cup (A - B) = A$$

Every element of A is either even (and thus in B) or odd (and thus in $A - B$). These are the only two cases, so $B \cup (A - B) = A$. ✓

Conclusion: Since all three conditions hold, $\mathcal{H} = \{B, (A - B)\}$ forms a valid partition of A . ■

Question 4 (30 Marks):

Q4.(a) Prove by Mathematical Induction that for all integers $n \geq 1$: (20 Marks)

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Basis Step:

(3 Marks)

Show the formula holds for $n = 1$:

- **LHS:** $1^3 = 1$
- **RHS:** $\left(\frac{1 \cdot (1+1)}{2} \right)^2 = \left(\frac{2}{2} \right)^2 = 1^2 = 1$

Since LHS = RHS = 1, the basis step holds. ✓

Inductive Hypothesis:

(2 Marks)

Assume the formula holds for some arbitrary integer $k \geq 1$, i.e., assume:

$$P(k): \quad 1^3 + 2^3 + 3^3 + \cdots + k^3 = \left(\frac{k(k+1)}{2} \right)^2$$

Inductive Step:

(15 Marks)

We must prove $P(k+1)$, i.e., show that:

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2} \right)^2$$

Step 1 — Start from the LHS of $P(k+1)$ and apply the inductive hypothesis:

$$\underbrace{1^3 + 2^3 + \cdots + k^3}_{\text{use IH}} + (k+1)^3 = \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3$$

Step 2 — Expand:

$$= \frac{k^2(k+1)^2}{4} + (k+1)^3$$

Step 3 — Factor out $(k+1)^2$:

$$\begin{aligned}
 &= (k+1)^2 \left[\frac{k^2}{4} + (k+1) \right] \\
 &= (k+1)^2 \left[\frac{k^2 + 4(k+1)}{4} \right] \\
 &= (k+1)^2 \cdot \frac{k^2 + 4k + 4}{4}
 \end{aligned}$$

Step 4 — Recognize the perfect square in the numerator:

$$k^2 + 4k + 4 = (k+2)^2$$

Step 5 — Substitute:

$$\begin{aligned}
 &= (k+1)^2 \cdot \frac{(k+2)^2}{4} \\
 &= \frac{(k+1)^2 (k+2)^2}{4} \\
 &= \left(\frac{(k+1)(k+2)}{2} \right)^2
 \end{aligned}$$

This is exactly the RHS of $P(k+1)$. ✓

Conclusion:

By the Principle of Mathematical Induction, since the basis step holds and the inductive step is proven, the formula

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

holds for all integers $n \geq 1$. ■

Q4.(b) Prove the set identity algebraically:

(10 Marks)

$$(A \cup B) \cap (A \cup B^c) = A$$

We apply standard Set Identity laws step by step:

$$\begin{aligned}
 (A \cup B) \cap (A \cup B^c) &= A \cup (B \cap B^c) && \text{(Distributive Law)} \\
 &= A \cup \emptyset && \text{(Complement Law: } B \cap B^c = \emptyset) \\
 &= A && \text{(Identity Law: } A \cup \emptyset = A)
 \end{aligned}$$

Laws Used (Summary):

Step	Law Applied
Step 1	Distributive Law: $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$ applied in reverse: $(A \cup B) \cap (A \cup B^c) = A \cup (B \cap B^c)$
Step 2	Complement Law: $B \cap B^c = \emptyset$
Step 3	Identity Law: $A \cup \emptyset = A$